

Inc) into polyethylene tubing (PE90, Becton Dickinson and Company) that was filled with pre-epoxy (5 minute Epoxy, Devcon). The use of polyethylene tubing as the insulating layer resulted in channels that were wide enough for us to insert glass capillaries directly, to serve as inlet/outlet channels. Holes for the inlets for the continuous phase were drilled with a needle ($60^{1/2}$, Becton Dickinson and Company), and polyethylene tubing (PE60) was inserted into the holes. Syringe pumps (PHD2000, Harvard Apparatus) were used to control the flow of fluids.

Images of nylon-coated droplets were collected with a charge-coupled device (CCD) camera (DMX1200, Nikon) connected to a stereo- or inverted microscope. The diameter of the droplets was measured by analyzing the images (Scion Image, Scion Corporation).

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Transport Properties in the Rubrene Crystal: Electronic Coupling and Vibrational Reorganization Energy**

By Demétrio A. da Silva Filho, Eung-Gun Kim, and Jean-Luc Brédas*

Over the past few years, pentacene has been viewed as the benchmark material for hole mobilities in organic semiconductors, with room-temperature mobilities as high as $35 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ being reported recently.^[1] In spite of remarkable progress, a number of issues still remain unclear. For instance, it is still not known what the highest carrier mobility that can be achieved in organic materials is, and what the chemical structures and packing modes that will trigger high mobility are.^[2] In this context, the investigation of molecular single crystals can provide a valuable source of information for understanding the microscopic parameters that determine transport efficiency.

Rubrene, a tetraphenyl derivative of tetracene, has lately been gaining much attention.^[3–7] Due to the bulky phenyl substituents attached to the side of the tetracene backbone (see Fig. 1), weak intermolecular interactions, and as a result small carrier mobilities, are a priori expected in this system. However, most interestingly, very recent measurements on ultrapure single crystals indicate a field-induced hole mobility of up to $20 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ at room temperature.^[7] Therefore, in the present work, we use quantum-chemical calculations of the electronic (band) structure and vibrational reorganization energy of rubrene to elucidate the origin of these large hole mobilities, and of their marked anisotropy along the crystallographic axes;^[7] comparison is also made to tetracene and pentacene.

In ultrapure single crystals at low temperature, the electronic coupling between adjacent molecules (that is, the interchain transfer integral, t) is the key parameter in describing the band-like, intrinsic transport properties;^[8–10] knowledge of the

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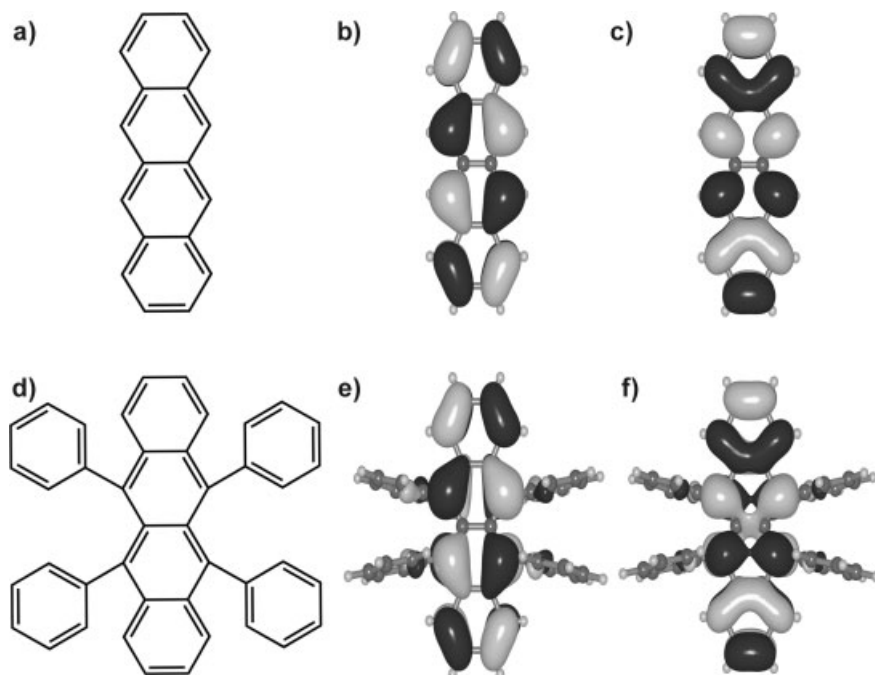


Figure 1. Chemical structures of a) tetracene and d) rubrene. DFT-B3LYP/6-31G(d,p)-calculated HOMO and LUMO wavefunctions in the neutral ground-state geometry: b) tetracene HOMO (−4.87 eV); c) tetracene LUMO (−2.09 eV); e) rubrene HOMO (−4.69 eV); and f) rubrene LUMO (−2.09 eV).

transfer integral is also important in evaluating the carrier mobilities in the (activated) hopping régime.^[9–13] Here, as we have done previously,^[10] the interchain transfer integrals for holes and electrons are first computed at the semiempirical intermediate neglect of differential overlap (INDO) level, by considering the splittings of the highest occupied and lowest unoccupied molecular orbital (HOMO and LUMO, respectively) levels in stacks of molecules built along the main crystallographic directions. We point out that in the isolated (neutral) molecules, the shapes and energies of the HOMO and LUMO levels of rubrene and tetracene are very similar, as seen in Figure 1; this reflects the large torsion angle between the phenyl rings and the tetracene backbone in rubrene (calculated in the isolated molecule and measured in the crystal^[14] to be $\sim 85^\circ$), which strongly reduces any mixing of the molecular orbitals between the backbone and the side groups.

The widths of the HOMO and LUMO bands, calculated at the INDO level, are plotted in Figure 2 as a function of $\cos(\pi/(N+1))$, where N is the number of rubrene molecules in the stacks built along the a and d directions (which are defined in the inset of Fig. 2 and in Fig. 3). Large transfer integrals are found along the a direction and lead to bandwidths, W , of ca. 340 and 160 meV for holes and electrons, respectively. Smaller interactions are calculated along the diagonal directions (d) while all other directions had negligible transfer integrals. Interestingly, the highest bandwidths in rubrene (along the a direction) are comparable to the highest bandwidths calculated in pentacene (along the d direction).^[10,15] We note that the linear evolution of the calculated bandwidths (Fig. 2) as a function of $\cos(\pi/(N+1))$ implies that the intermolecular interactions in rubrene can be cast into a tight-binding formalism, with $W = 4t$, as was also observed, for

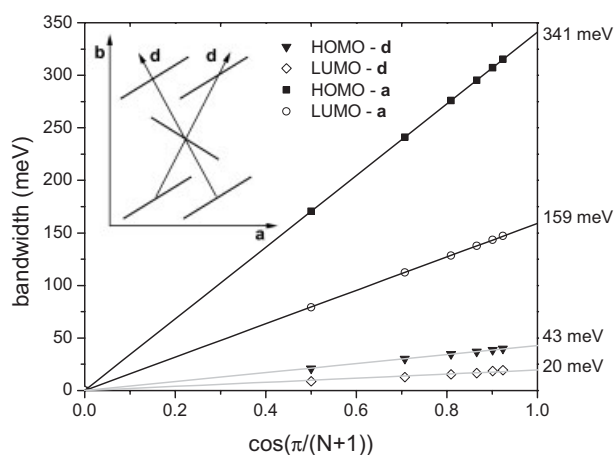


Figure 2. Evolution, as a function of $\cos(\pi/(N+1))$, of the INDO-calculated bandwidths formed by the HOMO and LUMO levels in rubrene stacks along the a and d directions. N is the number of interacting molecules. The inset illustrates the arrangement of the molecules in the herringbone ab plane. A linear fit was used to extrapolate the full bandwidth along each stack. The bandwidths along other directions are vanishingly small and are not reported.

example, for oligoacene, sexithiophene, and bisdithienothiophene crystals.^[15–17]

To understand the magnitude of the transfer-integral values, we need to examine the packing of the molecules in the crystal structure. Both rubrene and pentacene (or tetracene) present a herringbone motif in the ab plane where the most significant electronic couplings are found. However, there are major differences that are triggered by the presence of the phenyl side-groups in rubrene:

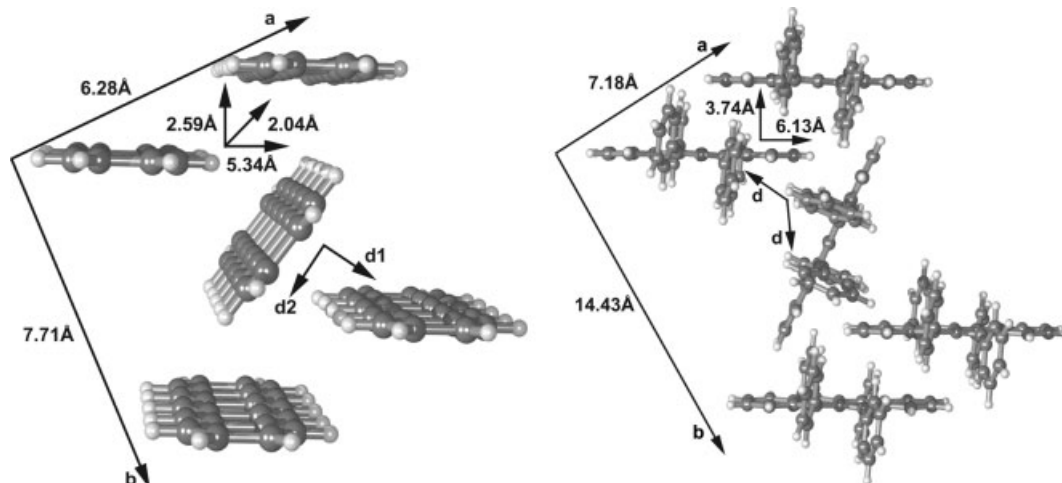


Figure 3. Illustration of the lattice parameters within the ab layer of crystalline rubrene (right) and pentacene (left). The short-axis and long-axis displacements along the π -stacks in the a direction are also indicated.

i) The long molecular axes all come out of the ab plane in pentacene (or tetracene) while, in rubrene, they are embedded in that plane (see Fig. 3). As a consequence, the long molecular axes of adjacent molecules along the diagonal (herringbone) directions are parallel in pentacene, while they are almost perpendicular in rubrene. This explains the smaller transfer integrals along the diagonal directions in rubrene.

ii) Along the crystal a direction, for which the highest bandwidths are calculated, the rubrene molecules are found to form a π -stack with a stacking distance of 3.74 Å; although this distance is larger than in a typical π -stack,^[18,19] a major feature is that no displacement occurs along the short molecular axes. In pentacene (or tetracene), the short-axis displacements are so large that adjacent molecules along a interact weakly, in spite of a very short π -stack distance of 2.59 Å (see Fig. 3).

While no short-axis displacement occurs along the a direction in rubrene, the phenyl side groups facilitate a very large sliding, of 6.13 Å, of one molecule with respect to the next along the long molecular axis (see Fig. 3). This leads to the appearance of the slipped-cofacial configuration illustrated in Figure 3. Such long-axis displacements are known to reduce the electronic coupling between adjacent molecules (which is maximal for the perfectly cofacial situation, oscillates between positive and negative values as a function of increasing displacement, and vanishes for large displacements^[9,10]). Thus, the question that needs to be answered at this stage, is how such a pronounced long-axis sliding can be consistent with the large bandwidths that are calculated along the a direction of the rubrene crystal. Therefore, we examined, at the INDO level, the evolution of the HOMO and LUMO transfer integrals for a complex made of two rubrene molecules. We started from a perfectly cofacial situation, with the two molecules superimposed at a distance of 3.74 Å, and then increasingly displaced the top molecule along the long molecular axis. Note that we actually replaced the phenyl side groups present in rubrene by hydrogen atoms (thereby effectively considering two

tetracene molecules), since the phenyls do not play any significant role in the HOMO or LUMO wavefunctions of rubrene (see Fig. 1) and their presence would obviously lead to major steric interactions upon displacement. The results are presented in Figure 4. The typical oscillating evolution of the HOMO and LUMO transfer integrals is found to occur upon displacement. Remarkably, the displacement of 6.13 Å observed in the rubrene crystal closely corresponds to extrema in the oscillations of both the HOMO and LUMO transfer integrals. This result clarifies why, even at such a large long-axis displacement, the transfer integrals are still a significant fraction of the values found for the cofacial case.

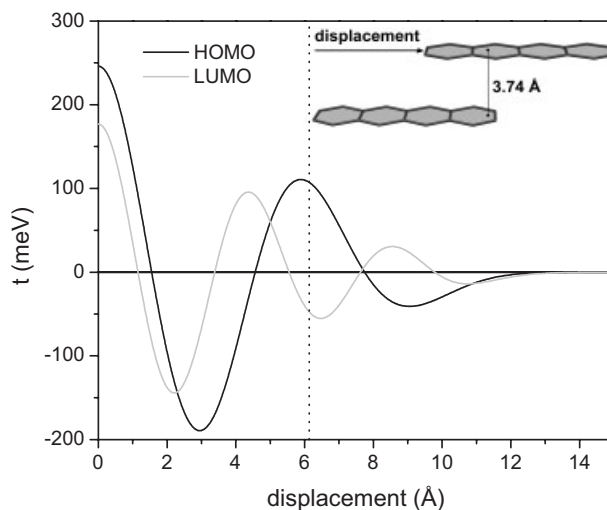


Figure 4. Evolution of the INDO HOMO and LUMO transfer integrals as a function of displacement, for a complex made of two tetracene molecules stacked along the rubrene a direction with a π -stacking distance of 3.74 Å. The dotted line indicates the magnitude of the long-axis displacement, 6.13 Å, found in the rubrene crystal. The molecular geometries were optimized at the DFT-B3LYP/6-31G(d,p) level.

Along the d direction of rubrene, the calculated HOMO and LUMO bandwidths are much smaller than along the a direction, in agreement with the experimental findings^[6] that show a strong anisotropy of the mobility within the herringbone planes of rubrene single crystals. We note that two adjacent molecules in the b direction are 14.4 Å apart (see Fig. 3); for such a distance, there is no electronic overlap between the two oligomers. As a result, a charge carrier moving along the b direction is expected to have to make its way via the intercalated molecules (in the d direction illustrated in Fig. 3); thus, the transfer integrals along the d direction should be used to understand the transport in the b direction. This argument is confirmed by the three-dimensional electronic band structure of the rubrene crystal calculated using the plane-wave density functional theory (PWDFT) method (see Fig. 5). Here, the π - π coupling across different orientations between the a and b axes, represented by the $\Delta_a\Theta Z$ direction in the Brillouin zone, leads to bands nearly as wide as those along the $\Gamma\Delta_a$ direction (corresponding to the a axis).

The PWDFT results also confirm that the rubrene valence bandwidth is of the order of a few tenths of an electron volt (ca. 0.4 eV). Thus, the INDO and DFT calculations concur^[21] to suggest that, in the absence of traps, a band régime should be present in rubrene single crystals at low temperature and could still be operative for holes at room temperature; the latter is consistent with the decrease in mobility with temperature observed around room temperature.^[6,7]

To achieve a more complete understanding of the charge-transport properties and to encompass the hopping régime that could appear at high temperature, we also calculated, at the DFT level, the intramolecular reorganization energy for holes in rubrene, and compared the results to those for holes in tetracene and pentacene.^[22] The reorganization energy is a key microscopic parameter that determines the rate of the activated hopping processes.^[9] The total intramolecular reorganization energy consists of two terms that reflect the geometry relaxation energies of one molecule (which accepts the hole) going from the neutral-state geometry to the positively charged-state geometry and an adjacent molecule (which donates the hole) evolving in the opposite way. The calculated total intramolecular reorganization energy for an isolated molecule of rubrene is 159 meV, which is nearly 40 % larger than that of tetracene (114 meV) and 70 % larger than that of pentacene (95 meV);^[22] we note, however, that this value remains significantly smaller than that of the bistriarylamine derivative tetraphenyldiaminobiphenyl (TPD) (290 meV),^[23] a well-established hole-transport material.

In order to understand the origin of the difference between rubrene and tetracene, the cationic relaxation energy was decomposed into contributions from individual vibrational modes (performed with the DUSHIN program developed by Reimers^[24]). Figure 6 shows the decompositions for tetracene and rubrene plotted against the cation normal modes. It appears clearly that the main difference between rubrene and tetracene comes from the contributions of low-frequency modes. While modes of frequency less than 1000 cm⁻¹ do not

contribute to the reorganization energy in tetracene, they contribute almost 50 meV to that of rubrene; this basically accounts for the difference in the intramolecular reorganization energy between the two systems. The major low-frequency vibrational mode (at 24 cm⁻¹) involved in the cationic relaxation energy in rubrene is shown in Figure 6; it mainly corresponds to a bending of the phenyl side-groups around the tetracene backbone. It is important, however, to bear in mind that this vibrational analysis was carried out on isolated oligo-

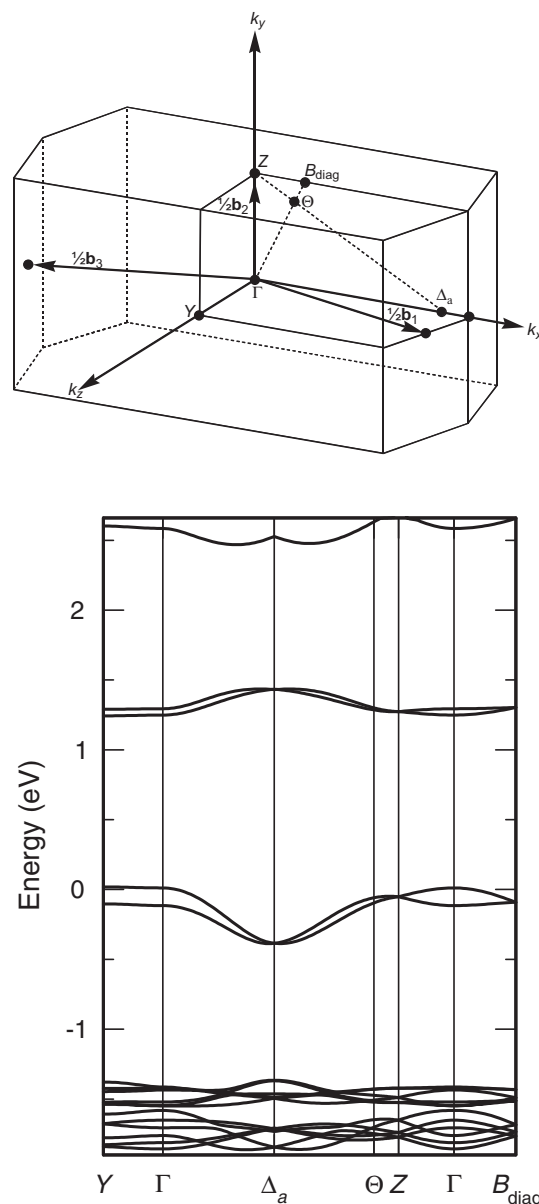


Figure 5. Band structure of the rubrene crystal, as obtained from a PWDFT calculation [20]. The calculation was carried out on the crystallographic geometry [14] within the generalized gradient approximation (BLYP) with Troullier-Martíns pseudopotentials. $\Gamma = 0$, $Y = (1/2)c^*$, $Z = (1/2)b^*$, and $\Delta_a = (1/2)a^*$. B_{diag} and Θ both have a diagonal orientation of $a + b$, with the former lying on the first Brillouin zone boundary and the latter corresponding to $(1/2)(1/|a + b|)$.

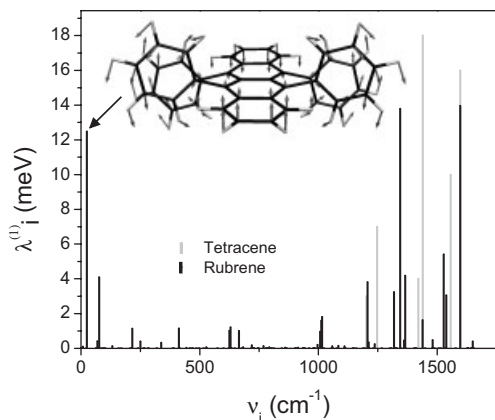


Figure 6. Contribution of the vibrational modes of rubrene and tetracene to the cationic relaxation energy, as calculated at the DFT/UB3LYP level with the 6-31G(d,p) basis set. Tetracene results were taken from a previous publication [21].

mers; these side-group bending vibrations might be hindered to some extent in the solid state and thus might turn out to play a lesser role in the hopping process.

To summarize, the results of our theoretical work indicate that, in spite of the presence of bulky side-groups, the electronic couplings and bandwidths in rubrene are comparable to those calculated for tetracene and pentacene. The origin of the large electronic coupling along the crystallographic *a* direction of rubrene is related to: i) the absence of short-axis displacements between adjacent molecules along *a* stacks; and ii) the presence of long-axis displacements that coincide with extrema in the evolution of the transfer integrals upon sliding. Our results suggest the existence, in the absence of traps, of a band régime for holes up to room temperature, with significant anisotropy of the hole mobility within the herringbone plane; these results are consistent with available experimental data.^[5–7] With regard to the hopping régime, our calculations show that the reorganization energy of rubrene is expected to be larger than in pentacene and tetracene, due to low-frequency motions of the phenyl side groups.

Our results confirm that cofacial π -stack interactions, such as those found along the *a* direction in rubrene, result in very efficient electronic couplings. Thus, the design of oligomers that pack in a way that favors this type of interaction^[18,25] is a promising route to increasing carrier mobilities in organic semiconductors.

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